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REV STATUS OF SHEETS											
REV	-	-	-	-	-	-	-	-	-	-	
SHEET	-	-	-	-	-	-	-	-	-	-	

DWG NO: <ASSY DWG NO> REV: .lfe

REVISIONS

ZONE	REV	DESCRIPTION	DATE	APPROVED
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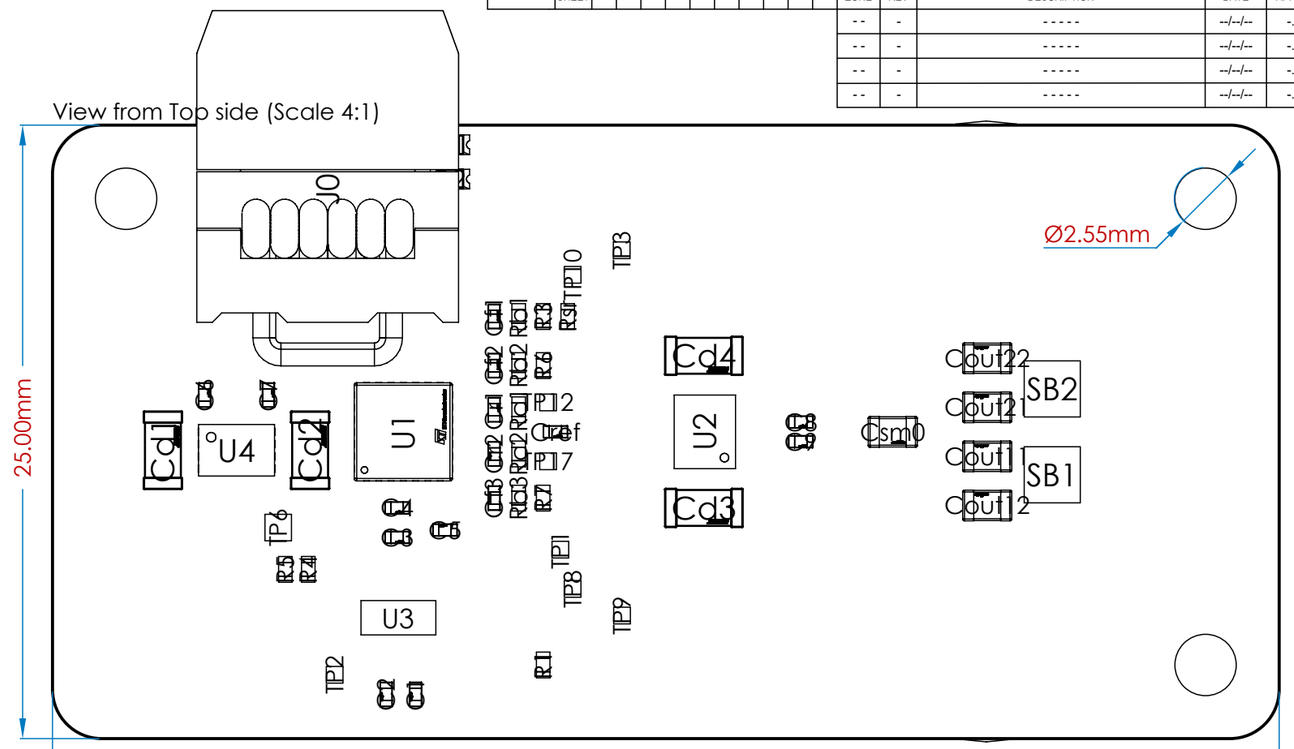
Bill Of Materials

Designator	Name	Quantity	JLCPCB Part #
U1	STM32G071GBU6	1	C529347
U3	MCP1799T-3302H/TT	1	C7424770
X1	RFM-0505S	1	
U2	MP6551GQB-P	1	C3681210
Lf1, Lf2	RCH1216BNP-100M	2	
U4	ISO6521REUR	1	C22428095
Cout11, Cout12, Cout21, Cout22, Csm0	GRM21BR61H475ME51L	5	C98192
D1, D2	150060RS75000	2	C2286
Csm1, Csm2	860010474012	2	
Cd1, Cd2, Cd3, Cd4	C1206C104M5RACTU	4	C24497
C1, C2, C3, C4, C5, C6, C7, C8, C9, Cref	GRM155R61E105KA12D	10	C52923
R1, R3, R4, R5, R6, R7, Rsr	RC0402FR-0710KL	7	C25744
Rb1, Rb2, Rb3, Rc1, Rc2, Rd1, Rd2	RC0402FR-073K3L	7	C25890
Cr1, Cr2	GJM1555C1H100JB01D	2	C32949
Cf1, Cf2, Cf3	CL05B103KB5NNNC	3	C15195
SB1, SB2	Solder-Bridge-NO	2	

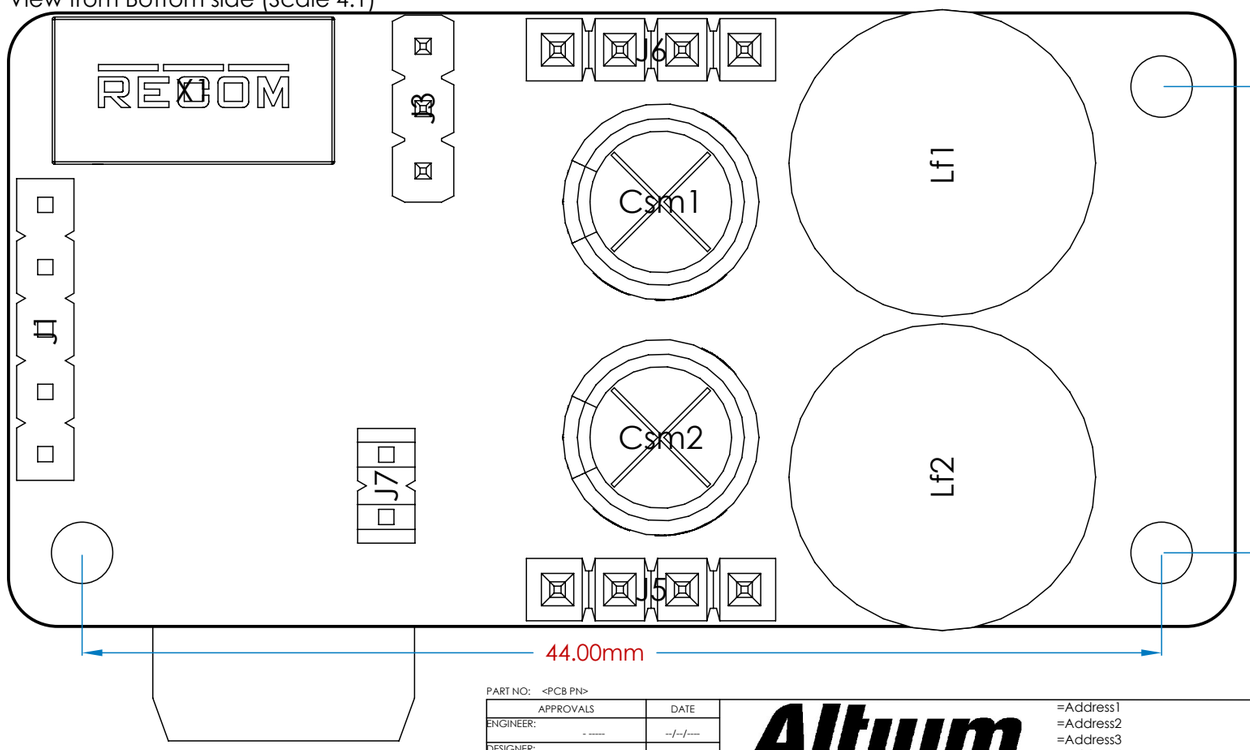
MCU Pinout

Pin (Nb) Name	Pin Mode	Function	Notes
Motor Drive Control			
(16) PA8	TIM1 CH1	PWM1	PWM input of Bridge 1
(18) PA9	TIM1 CH2	PWM2	PWM input of Bridge 2
(7) PA1	Output	EN1	Enable input of Bridge 1
(17) PC6	Output	EN2	Enable input of Bridge 2
(2) PC15	Input	nFAULT	Use internal pull-up
Timer Synchronisation			
(15) PB1	TIM1 CH3n	TIM1 SYNC	Connected together. Master: CH3n, Slave: ETR
(19) PA12	TIM1 ETR	TIM1 SYNC	
(23) PB3	TIM2 CH2	TIM2 SYNC	Connected together. Master: CH2, Slave: ETR
(22) PA15	TIM2 ETR	TIM2 SYNC	
User Interface			
(24) PB4	TIM3 CH1	LED STAT	LED controlled in software about Status
(25) PB5	TIM3 CH2	LED COM	LED controlled in software about Communication
(10) PA4	DAC1 OUT1	TP	Programmable test pin DAC or TIM14 CH1
Measurement			
(11) PA5	DAC1 OUT2	Vref	Reference voltage for current measurement
(09) PA3	ADC1 IN3	CM1	Low-side current measurement of Bridge 1
(12) PA6	ADC1 IN6	CM2	Low-side current measurement of Bridge 2
(08) PA2	ADC1 IN2	Vout1	Output voltage measurement of Bridge 1
(13) PA7	ADC1 IN7	Vout1	Output voltage measurement of Bridge 2
(6) PA0	ADC1 IN0	Vref	Measurement of reference voltage
(14) PB0	ADC1 IN8	Vsm	Measurement of SM voltage
(1) PC14	Input	Viso	Binary detection of external supply
Communication			
(26) PB6	USART1 TX	UART	Serial transmission
(27) PB7	USART1 RX	UART	Serial reception
MCU Programmation			
(20) PA13	SYS SWCLK	Prog	Programmation pin Clock
(21) PA14	SYS SWDIO	Prog	Programmation pin Data
(5) PF2	Input	nRESET	Reset pin

View from Top side (Scale 4:1)



View from Bottom side (Scale 4:1)



PART NO: <PCB PN>

APPROVALS		DATE
ENGINEER:	- -----	--/--/----
DESIGNER:	- -----	--/--/----
CHECKER:	- -----	--/--/----

Reference Documents

BOM DOC:	<BOM DOC NO>
ASSY DOC:	<FAB DWG NO>
SCH DOC:	<SCH DWG NO>
PCB DOC:	<PCB DWG NO>

THIRD ANGLE PROJECTION

NEXT ASSY | USED ON

APPLICATION

Altium™

=Address1
=Address2
=Address3
=Address4

DESIGN ITEM: .Item | DESIGN ITEM REVISION: .ItemRevision

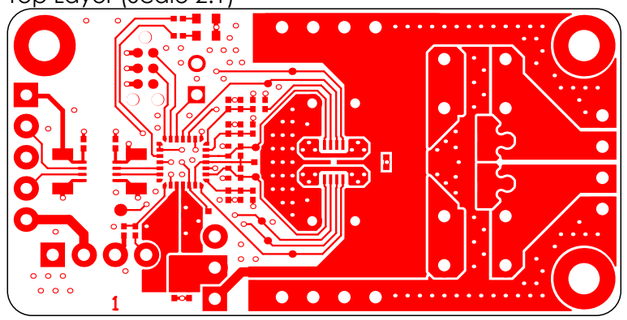
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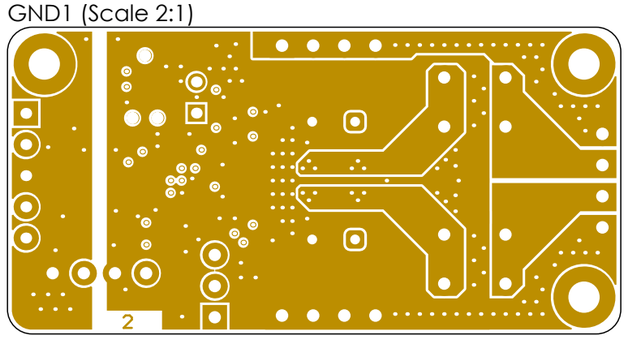
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DWG NO: <ASSY DWG NO> REV: .lfe

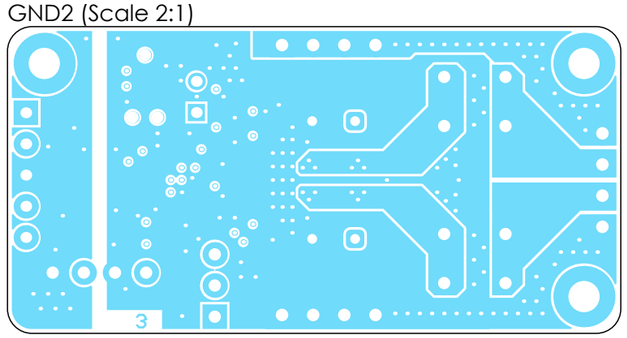
Top Layer (Scale 2:1)



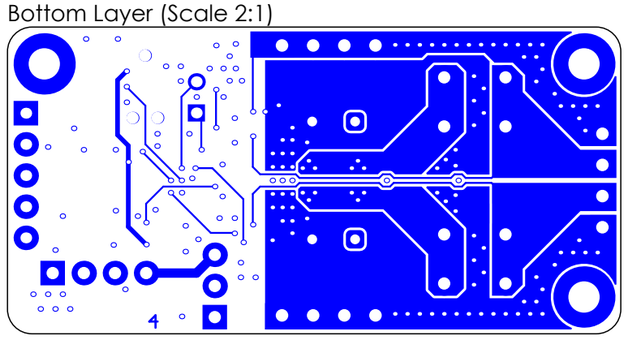
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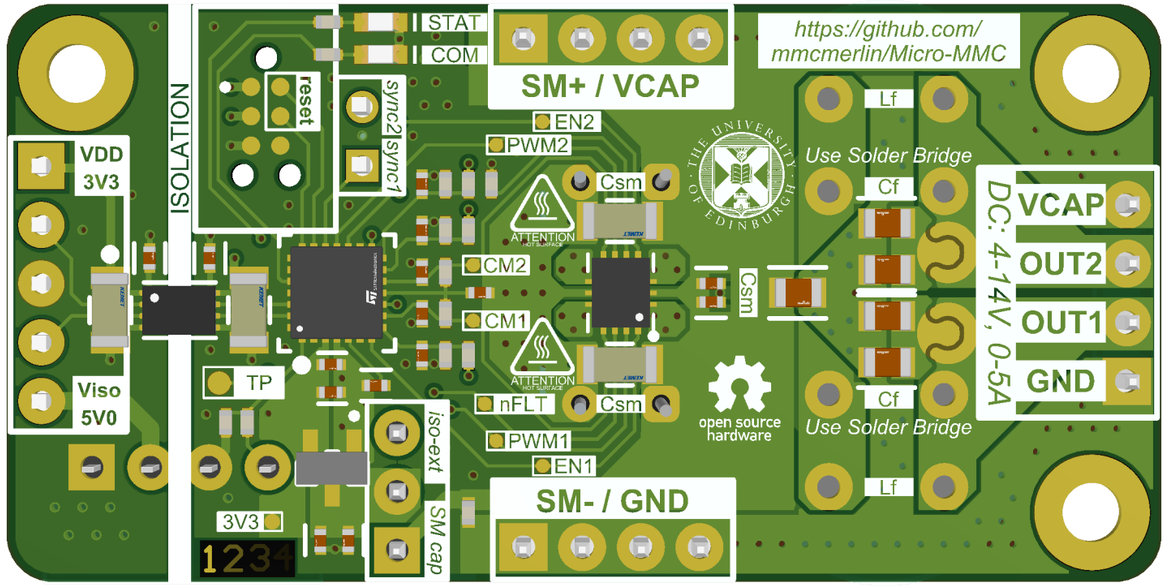
GND2 (Scale 2:1)



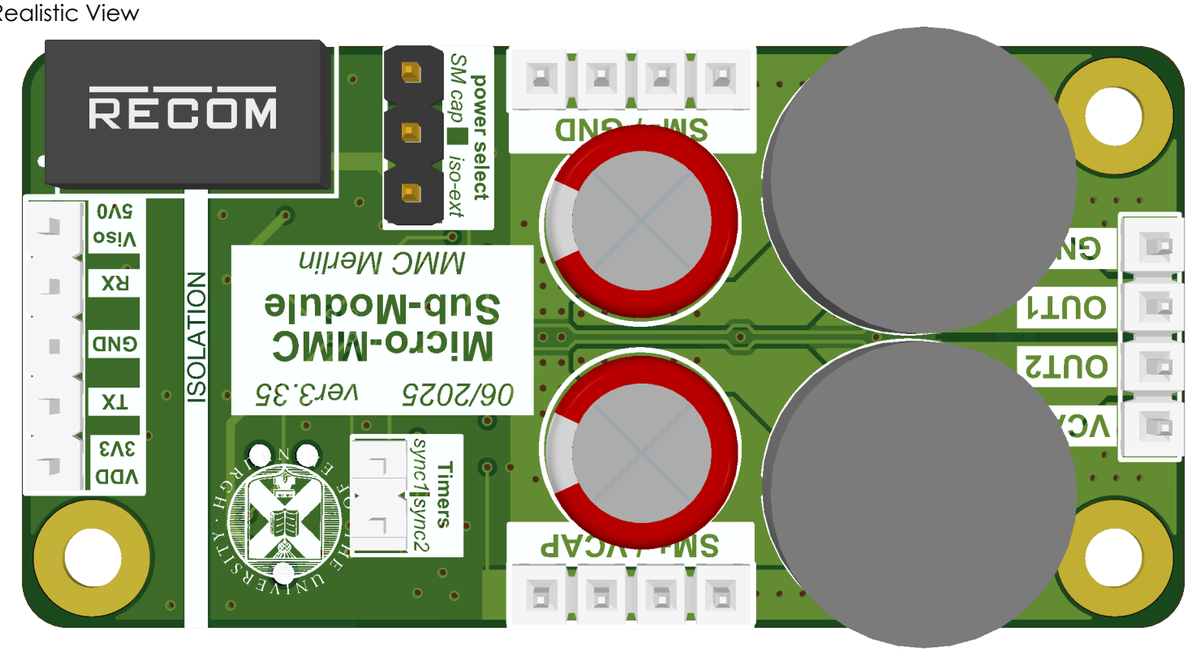
Bottom Layer (Scale 2:1)



Realistic View



Realistic View



SIZE:	CAGE CODE:	DWG NO:	REV:
A3	?????	<FAB DWG NO>	
SCALE:	FILE NAME: MicroMMC_SM_v3.PCBDwf	SHEET: 2 OF 2	

DWG NO: <ASSY DWG NO>
REV: .life